APURV DAS

Software Engineer — C++20 · Windows · Direct3D 11 · Real-Time Audio & Graphics

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SUMMARY

Software engineer (MCA, 2026) who builds native software that stays fast on weak hardware. Rewrote a DJ-library app as a single native C++20 / Win32 / Direct2D executable (first paint in about 0.1 s, 0% idle CPU, about 43 MB of RAM) and built DasVis, a Direct3D 11 music visualizer that analyses each song's structure and runs at about 5% of one CPU core on GPUs back to feature level 10_0. Most at home where software meets devices: low-latency audio clocks over WASAPI, GPU timing and quality tiers, Windows device and radio APIs, and network protocols implemented byte for byte. Everything I ship has tests and CI.

TECHNICAL SKILLS

- **Languages:** C++20, Python, JavaScript, TypeScript, PHP, SQL, HLSL (Shader Model 4), HTML5, CSS3, Rust (basic, Tauri)
- **Windows & Systems:** Win32, COM, Direct2D / DirectWrite, Direct3D 11 / DXGI, WASAPI (miniaudio), Media Foundation, WIC, WinHTTP, CNG, SetupAPI, WLAN API, WinRT Bluetooth, multithreading
- **Audio, Video & Graphics:** FFT / DSP (own SSE2 FFT), STFT, harmonic-percussive separation, beat tracking, BPM and key detection, HDR rendering and bloom, GPU fluid and reaction-diffusion simulation, GPU timestamp profiling, H.264 encoding, libVLC, projectM, TagLib
- **Test Tools:** Playwright, pytest, PHPUnit, Vitest, CTest, unittest, JIRA, Android Studio, Android Debug Bridge (ADB), Android Emulator, Chrome DevTools
- **Networking & Security:** TCP / UDP sockets, HTTP servers and clients, binary protocol implementation, REST APIs, OAuth 2.0 (PKCE, Google OAuth), JWT, nmap, XSS / CSRF / IDOR testing
- **DevOps & Tools:** Git, GitHub Actions (CI/CD), GitHub Pages, CMake, vcpkg, MSVC, PowerShell, Docker, AWS

PROJECTS

WreckBox for Windows — Native C++ Desktop App — Personal Project | Oct 2026

Tools: C++20, Win32, Direct2D, libVLC, miniaudio (WASAPI), projectM, TagLib, CMake / vcpkg, CTest, PowerShell, GitHub Actions (a native rewrite of an existing DJ-library app) | Download: github.com/moloyb301-eng/wreckbox-releases

- Rewrote the app as one native executable: first paint in 105–147 ms (300 ms budget), search across 5,000 tracks in 0.8 ms, 0% idle CPU (no frame loop), about 43 MB of RAM at rest.
- Halved memory after scrolling (about 92 to 50 MB) by fixing GPU brush retention and moving to the Windows segment heap.
- Ported the Rust audio-analysis engine (tempo, key, energy) to C++ with my own SSE2 FFT, computed frame by frame so a 5-minute track needs no spectrogram in memory; results match the Rust engine on every file both can read.
- Built a low-latency playback path (libVLC into a 0.25 s ring, miniaudio / WASAPI at 20 ms periods) with a sample-accurate "heard" clock that keeps visuals on the beat, including automatic Bluetooth latency compensation.
- Implemented the Soulseek peer-to-peer protocol natively (server, peers, relayed transfers), with 28 messages verified byte for byte against a reference implementation.
- Wrote 12 CTest suites (690+ assertions) and configured CI that builds, tests, smoke-tests and packages each release; smoke testing caught a crash on every app exit (2 root causes fixed).

DasVis — Direct3D 11 Music Visualizer — Personal Project | Oct 2026 – Present

Tools: C++20, Direct3D 11 (feature level 10_0, Shader Model 4), HLSL, Media Foundation | Code: github.com/ApurvDas/dasvis

- Streaming song analysis (STFT, harmonic-percussive separation, dynamic-programming beat tracking, novelty-based section detection) maps a song's beats, build-ups and drops in about 4–6 s using about 6 MB, so the visuals anticipate drops.
- 15 GPU scenes, including stable-fluids, reaction-diffusion and a one-million-star galaxy, all as pixel shaders on ping-pong textures; quality tier and render scale chosen from GPU timestamp queries; about 5% of one CPU core and +20 MB of RAM at the low tier.
- Photosensitivity-safe flash limiter (at most 3 flashes a second), frame-exact H.264 video export through Media Foundation, Spout GPU texture sharing, and WLED LED-strip control over realtime UDP.
- Automated GPU rendering tests on a software renderer (WARP), including determinism tests (same input, bit-identical frames) and a photosensitivity (accessibility) test.

Unigen Manager — Windows Wi-Fi & Bluetooth Manager — Personal Project | Oct 2026

Tools: Python, ctypes (SetupAPI, WLAN API), WinRT, pywebview + WebView2, anime.js, pytest, Playwright | Code: github.com/ApurvDas/unigen-manager

- Manages a USB Wi-Fi 6 + Bluetooth dongle and a laptop's built-in card side by side from one dark-mode window with live charts and an always-on tray widget; ships as one 33 MB executable with no installer.
- Moves Wi-Fi and Bluetooth between the dongle and the built-in card by switching radios and devices through SetupAPI and the WLAN API (Wi-Fi moves in about 3 s with no driver restart), with an unplug / replug watchdog and rollback on failure; reads device battery levels from Windows device properties.
- Wrote 653 pytest tests (unit, API contract, Playwright UI tests in Microsoft Edge) plus 20 read-only hardware checks on the real adapters and a 24-case manual test checklist for system-changing actions.

Project Manager — Offline-First Web App + Windows Desktop App — Personal Project | Aug 2026 – Present

Tools: Supabase/PostgreSQL, JavaScript, Tauri (Rust), Playwright, GitHub Actions | Code: github.com/ApurvDas/project-manager

- Live Supabase app with an offline-first sync engine (IndexedDB plus an outbox replayed through server RPCs) and a Tauri Windows desktop app using about 170 MB idle against about 315 MB for Chrome; 60 Playwright E2E tests and 32 API permission tests.

NetRisk AI — Home-Network Risk Assessment — Course Project | Oct 2026

Tools: Python, nmap, SQLite, Windows Task Scheduler

- Scans the top 200 TCP ports of every device on the home subnet every 6 hours, unattended on a 4 GB laptop, and scores risk with a per-device anomaly model; alerts only; 44 unit tests with mocks.

Other Projects — Personal & Academic | 2026

- Ciba Library (Next.js 16, React 19, Prisma, anime.js, react-three-fiber), Focus (FastAPI, SQLite), Khaata (Flask, Tkinter) and a one-file SmartThings A/C command-line tool in Python.

WORK EXPERIENCE

QA Tester & Product Contributor (Part-Time) — Zentrax Solutions LLP | Apr 2026 – Present

- Perform manual functional, regression, exploratory and UI/UX testing of an intercity cab-sharing platform: the Android app, the web admin panel (backoffice) and its backend APIs.

AI / ML Intern (Remote Mentoring Program) — Pheme Software Pvt. Ltd. (IBM Platinum Business Partner), Remote | Jan 2026 – May 2026

- Built "AI Visual Search Engine for E-Commerce", an image-based product search system using CNN feature extraction and image similarity matching in Python.

Web Developer Intern — Technotask Business Solutions, Bhopal | May 2023 – Jun 2024

- Developed and maintained web applications using PHP, CodeIgniter (MVC), HTML5, CSS3, JavaScript and MySQL.

EDUCATION

Master of Computer Applications (MCA) — UPES, Dehradun | Aug 2024 – Jul 2026 (Completed)

Bachelor of Computer Applications (BCA) — Christ (Deemed to be University), Delhi NCR | Apr 2024

Higher Secondary — NIOS | Jun 2020